Beam Current Limits

USPAS - Beam Physics with Intense Space Charge

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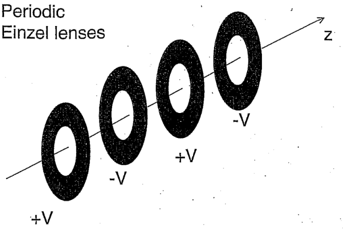


PERIODIC SOLENOIDS

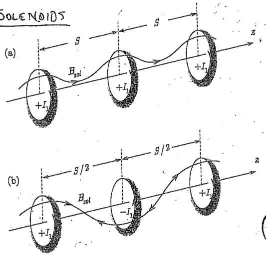


Figure 3.2. Schematic of magnet sets producing a periodic focusing solenoidal field with axial periodicity length S. In Fig. 3.2 (a), successive coils are spaced by S and have the same current polarity $+I_1, +I_1, \cdots$. In Fig. 3.2 (b), successive coils are spaced by $S/2$ and have alternating current polarities $+I_1, -I_1, +I_1, \cdots$.

(FIGURE FROM DAVIDSON & QIN, 2003) p. 55
"PHYSICS OF INTENSE CHARGED PARTICLE BEAMS IN HIGH ENERGY ACCELERATORS"

1.1 Radial envelope equations

$$\tilde{r}'' + \frac{(\gamma\beta)'}{(\gamma\beta)}\tilde{r}' + \left(\frac{\gamma''}{2\gamma\beta^2}\right)\tilde{r} + \left(\frac{\omega_c}{2\gamma\beta c}\right)^2\tilde{r} - \left(\frac{\langle p_\theta\rangle}{\gamma m\beta c}\right)^2\frac{1}{\tilde{r}^3} - \frac{\varepsilon_r^2}{\tilde{r}^3} - \frac{Q}{2\tilde{r}} = 0 \quad (1)$$

$$p_\theta \equiv \gamma m r^2\dot{\theta} + qrA_\theta, \quad (2)$$

$$\varepsilon_r \equiv \sqrt{\langle r^2\rangle\langle r'^2\rangle - \langle rr'\rangle^2 + \langle r^2\rangle\langle r^2\theta'^2\rangle - \langle r^2\theta'\rangle^2} \quad (3)$$

$$Q \equiv \frac{1}{mc^2\beta^2\gamma^3}\frac{q\langle\lambda(r)\rangle}{2\pi\epsilon_0} \quad (4)$$

1.2 Solenoid focusing

- Long solenoid: $\mathbf{B} = B_z \hat{z}$.
- No acceleration $\gamma' = \gamma'' = 0$.
- Set $p_\theta = 0$ and $\epsilon_r = 0$ to minimizing "repelling" forces.

$$\tilde{r}'' + \left(\frac{\omega_c}{2\gamma\beta c}\right)^2 \tilde{r} - \frac{Q}{2\tilde{r}} = 0 \tag{5}$$

Maximum beam perveance occurs at equilibrium, when $\tilde{r}' = \tilde{r}'' = 0$:

$$Q = \frac{1}{2}\left(\frac{\omega_c \tilde{r}}{\gamma\beta c}\right)^2 \tag{6}$$

1.3 Derivation from force balance in non-relativistic beams and the line goes longer

- Particle moves with angular velocity $\theta = \omega r$.
- Particle sees linear focusing force from solenoid field $B_z = \omega_c m/q$.
- Within beam envelope, particle sees linear defocusing force from space charge.

$$\underbrace{m\omega^2 r}_{\text{centrifugal}} + \underbrace{\frac{Qm\beta^2 c^2}{2\tilde{r}^2} r}_{\text{centrifugal}} = \underbrace{q v_\theta B_z}_{\text{magnetic}} \tag{7}$$

$$\omega^2 r + \frac{Q\beta^2 c^2}{2\tilde{r}^2} r = \omega_c \omega r \tag{8}$$

$$Q(\omega) = \frac{2\tilde{r}^2}{\beta^2 c^2}\left(\omega\omega_c - \omega^2\right) \tag{9}$$

$$\omega^* = \max_{\omega} Q(\omega) = \frac{\omega_c}{2} \tag{10}$$

$$Q(\omega^*) = \frac{1}{2}\left(\frac{\omega_c \tilde{r}}{\beta c}\right)^2 \tag{11}$$

Equivalent to result from envelope equations when $\gamma^2 \to 1$:

$$Q = \frac{1}{2}\left(\frac{\omega_c \tilde{r}}{\gamma\beta c}\right)^2 \tag{12}$$

1.4 Angular kick at solenoid entrance

Approximate solenoid field as step function:

$$B_z(z) = B_0 \left[\Theta(z) + \Theta(L - z) - 1\right] = \begin{cases} 0 & \text{if } z < 0 \\ B_0 & \text{if } 0 \leq z \leq L \\ 0 & \text{if } z > L \end{cases} \tag{13}$$

$$\frac{\partial B_z}{\partial z} = B_0 \left[\delta(z) - \delta(L - z)\right] \tag{14}$$

Use earlier result to relate longitudinal derivative to radial magnetic field:

$$B_r(r,z) \approx -\frac{r}{2}\frac{\partial B_z}{\partial z} = \frac{rB_0}{2}\left[\delta(z) - \delta(L-z)\right] \tag{15}$$

Compute change in θ component of momentum at $z = \epsilon$, just after solenoid entrance:

$$\frac{d}{dt}p_\theta^*(r) = qv_zB_r \tag{16}$$

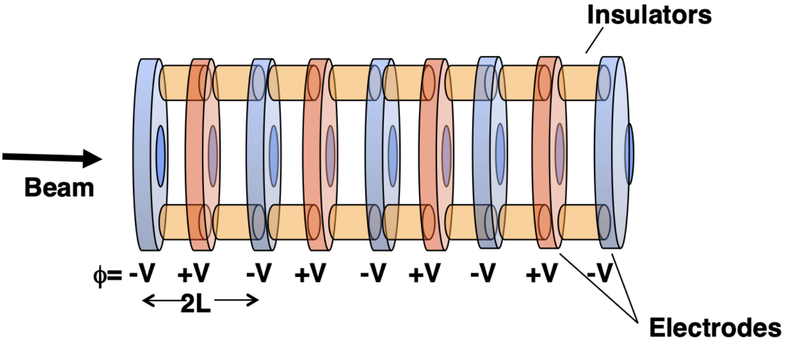
$$\Delta p_\theta^*(r) = q\int_{-\infty}^{\epsilon} B_r(r,z)dz \tag{17}$$

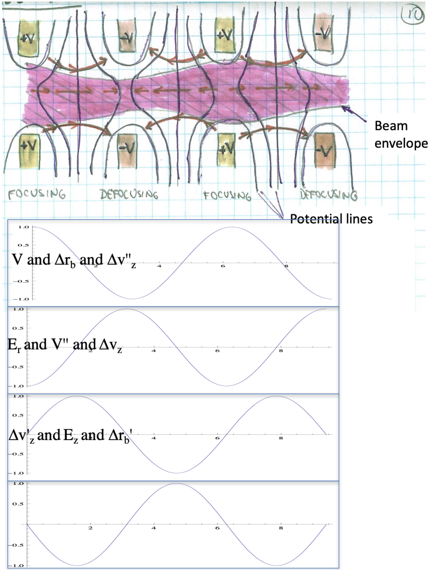
$$= \frac{qrB_0}{2}\int_{-\infty}^{\epsilon}\left[\delta(z) - \delta(L-z)\right]dz \tag{18}$$

$$= \frac{qrB_0}{2} \tag{19}$$

$$v_\theta = \frac{rqB_0}{2m} = \frac{r\omega_c}{2} \tag{20}$$

1.5 Einzel lens





$$V \Rightarrow v_z,\ V'' \Rightarrow E_r \Rightarrow \Delta r$$

Ignore centrifugal term, emittance term, and solenoid term (B_z) in radial envelope equation.

$$\tilde{r}'' + \frac{\gamma'}{\gamma\beta^2}\tilde{r}' + \frac{\gamma''}{2\gamma\beta^2} - \frac{Q}{2\tilde{r}} = 0 \tag{21}$$

Assume non-relativistic beam: $\beta \ll 1$, $\gamma \approx 1$:

$$\gamma' = \beta\beta'\gamma^3 \approx \beta\beta' \tag{22}$$

$$\gamma'' \approx \beta'\beta' + \beta\beta'' \tag{23}$$

$$\tilde{r}'' + \frac{\beta'}{\beta}\tilde{r}' + \frac{1}{2}\left[\frac{\beta'^2}{\beta^2} + \frac{\beta''}{\beta}\right]\tilde{r} - \frac{Q}{2\tilde{r}} = 0 \tag{24}$$

Define new variable $R = \sqrt{\beta/\beta_0}\tilde{r}$, where β_0 is the initial velocity.

$$\tilde{r}' = \left(\frac{\beta}{\beta_0}\right)^{-1/2} R' - \frac{1}{2}\left(\frac{\beta}{\beta_0}\right)^{-3/2}\left(\frac{\beta'}{\beta}\right) R \tag{25}$$

$$\tilde{r}'' = \left(\frac{\beta}{\beta_0}\right)^{-1/2} R'' - \left(\frac{\beta}{\beta_0}\right)^{-3/2}\left(\frac{\beta'}{\beta_0}\right) R' + \frac{3}{4}\left(\frac{\beta}{\beta_0}\right)^{-5/2}\left(\frac{\beta'}{\beta_0}\right)^2 R - \frac{1}{2}\left(\frac{\beta}{\beta_0}\right)^{-3/2} \tag{26}$$

$$\boxed{R'' + \frac{3}{4}\left(\frac{\beta'}{\beta}\right)^2 R - \frac{1}{2}\frac{\beta}{\beta_0}\frac{Q}{R} = 0.} \tag{27}$$

Approximate potential as $\phi(r,z) = \phi_0 \cos(\pi z/L)$. From conservation of energy:

$$\Delta\left(\frac{1}{2}mv^2\right) = -\Delta\left(\frac{q\phi}{m}\right) \tag{28}$$

$$\frac{1}{2}m\left(v^2 - v_0^2\right) = -q\phi_0 \cos\left(\frac{\pi z}{L}\right) \tag{29}$$

$$v^2 = v_0^2 - \frac{2q\phi_0}{m}\cos\left(\frac{\pi z}{L}\right) \tag{30}$$

Use this to rewrite $(\beta'/\beta)^2$:

$$v' = \frac{q\phi_0}{mv}\left(\frac{\pi}{L}\right)\sin\left(\frac{\pi z}{L}\right) \tag{31}$$

$$\frac{v'}{v} = \frac{q\phi_0}{mv^2}\left(\frac{\pi}{L}\right)\sin\left(\frac{\pi z}{L}\right) \tag{32}$$

$$\left(\frac{\beta'}{\beta}\right)^2 = \left(\frac{q\phi_0}{mv}\right)^2\left(\frac{\pi}{L}\right)^2\sin^2\left(\frac{\pi z}{L}\right) \tag{33}$$

Assume $v_0^2 \gg \frac{2q\phi_0}{m}$, so that $v \approx v_0$.

$$\left(\frac{\beta'}{\beta}\right)^2 \approx \left(\frac{q\phi_0}{mv_0}\right)^2 \left(\frac{\pi}{L}\right)^2 \sin^2\left(\frac{\pi z}{L}\right) \tag{34}$$

For equilibrium ($R'' = R' = 0$), look at averages over z:

$$\overline{\left(\frac{\beta'}{\beta}\right)^2} = \frac{1}{2}\left(\frac{q\phi_0}{mv_0}\right)^2 \left(\frac{\pi}{L}\right)^2 \tag{35}$$

$$\overline{\left(\frac{\beta}{\beta_0}\right)} = 0 \tag{36}$$

$$\overline{R} = \overline{\left(\frac{\beta}{\beta_0}\right)^{1/2}} \tilde{r} = \tilde{r} \tag{37}$$

Inserting these terms in the envelope equation with $R'' = 0$ gives the maximum beam perveance:

$$\boxed{Q_{max} = \frac{3\pi^2}{4}\left(\frac{q\phi_0}{m\beta_0^2 c^2}\right)^2 \left(\frac{\tilde{r}}{L}\right)^2} \tag{38}$$

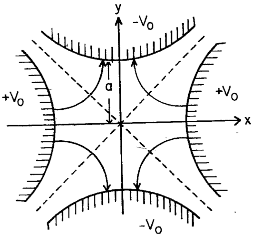
Outline

1. Beam current limits in axisymmetric focusing channels

2. Beam current limits in quadrupolar focusing channels

FROM
REISER, p.112

$E_x = -E'x$
$E_y = E'y$



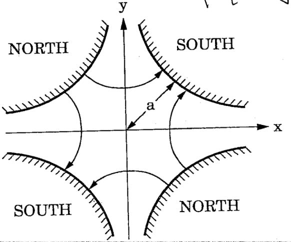
$F_x = -qE'x$
$F_y = qE'y$

ELECTROSTATIC
QUADS

Figure 3.15. Electrodes and force lines in an electrostatic quadrupole.

$B_x = B'y$
$B_y = B'x$

$F_x = -qV_z B'x$
$F_y = qV_z B'y$



MAGNETIC
QUADS

2.1 RMS envelope equations

$$\tilde{x}'' + \frac{(\gamma\beta)'}{(\gamma\beta)}\tilde{x}' + \kappa_x(s)\tilde{x} - \frac{\varepsilon_x(s)^2}{\tilde{x}^3} - \frac{Q}{2\,(\tilde{x}+\tilde{y})} = 0, \tag{39}$$

$$\tilde{y}'' + \frac{(\gamma\beta)'}{(\gamma\beta)}\tilde{y}' + \kappa_y(s)\tilde{y} - \frac{\varepsilon_y(s)^2}{\tilde{y}^3} - \frac{Q}{2\,(\tilde{x}+\tilde{y})} = 0. \tag{40}$$